

FINAL ML REPORT

Your model evaluation is complete.

A consolidated summary of dataset preparation, model performance, feature importance, and the final model recommendation.

RECOMMENDED MODEL

Random Forest

Random Forest achieved the strongest overall balance of predictive performance, class-sensitive evaluation metrics, and training efficiency among the evaluated models.

VALIDATION ACCURACY

86.4%

F1 Score

83.3%

ROC-AUC

89.1%

Training Time

2.8s

Models Evaluated

4

DATASET TRANSFORMATION

From raw data to training-ready dataset.

ORIGINAL DATASET

Records	10,000
Features	21
Missing values	3.2%
Duplicate rows	18

PROCESSED DATASET

Records	9,982
Features	23
Missing values	0%
Dataset status	Training ready

MODEL COMPARISON

Performance across evaluated models.

MODEL	ACCURACY	F1 SCORE	ROC-AUC	TRAINING	RANK
Random Forest	86.4%	83.3%	89.1%	2.8s	#1
XGBoost	85.8%	82.5%	88.4%	4.6s	#2
SVM	83.9%	80.6%	86.7%	3.4s	#3
Logistic Regression	81.7%	77.4%	84.0%	0.8s	#4

SELECTION REASONING

Why Random Forest was selected.

- ✓ **Strongest overall performance**
Achieved the highest validation accuracy and F1 score among the evaluated candidates.
- ✓ **High predictive separation**
The ROC-AUC score indicates strong ability to distinguish between the target classes.
- ✓ **Balanced training efficiency**
Delivered the strongest results without requiring the longest training time.
- ✓ **Suitable dataset fit**
Effectively captures nonlinear relationships and feature interactions within the dataset.

FEATURE IMPORTANCE

Features with the strongest influence.

Contract type, monthly charges, and customer tenure emerged as the most influential features in the final model evaluation.

Contract	21%
MonthlyCharges	17%
Tenure	14%
TotalCharges	11%
InternetService	8%

FINAL RECOMMENDATION

Random Forest is recommended as the primary model for this dataset.

Based on the comparative evaluation, Random Forest provides the strongest balance of predictive performance, F1 score, ROC-AUC, and training efficiency. XGBoost remains a close alternative, while Logistic Regression can serve as a faster and more interpretable baseline when computational simplicity is prioritized.

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